Weekly Record W5: Operational Process Engineering & Bottleneck Elimination at Atyrau Refinery

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Company A: Shell

Company B: KazMunayGas (KMG)

1. Academic & Theoretical Foundation

Operational process engineering in oil refining centers on maximizing throughput efficiency, energy integration, and equipment reliability while minimizing unplanned downtime. Grounded in the Theory of Constraints, every complex industrial system contains at least one bottleneck that limits overall system throughput. Eliminating operational bottlenecks requires systematic process mapping (Current State vs. Future State) and targeted technological interventions.

At complex refineries like the Atyrau Refinery (ANPZ - 5.5 million tonnes/year capacity), operational bottlenecks frequently emerge at the intersection of thermal processing units (catalytic cracking, hydrotreating) and heat exchange networks. Legacy manual inspection protocols create data latencies of 12 to 24 hours, delaying maintenance responses and leading to unscheduled unit trips.

2. Practical Artifact: Current State vs. Future State Process Flow at ANPZ

Table 5.1: ANPZ Heat Exchanger & Catalytic Cracking Process Optimization

Process Stage	Current State Workflow (Legacy)	Identified Bottleneck / Friction Point	Future State Workflow (AI-Enabled)	Target Operational Impact
1. Sensor Telemetry	Manual pressure & temperature gauge checks every 8 hours	Delayed detection of thermal fouling and vibration anomalies	Continuous IoT sensor streams (10-second intervals) to central server	Real-time visibility into equipment thermal efficiency

2. Data Processing	Manual data entry into Excel spreadsheets	12- 24 hour processing lag; high risk of human error	Automated ingestion into predictive analytics pipeline	Zero processing latency; instant trend detection
3. Anomaly Analysis	Weekly engineering review of historical logs	Reactive maintenance performed *after* failure occurs	ML algorithm predicts fouling 14 days prior to critical threshold	Shifts maintenance from reactive to proactive scheduled interventions
4. Maintenance Action	Emergency shutdown triggered by mechanical failure	Severe financial loss ($500k+/day) and domestic supply deficit	Scheduled bypass and cleaning during planned minor window	15% reduction in unplanned refinery downtime

Table 5.2: ANPZ Operational Control Metrics

Key Performance Indicator (KPI)	Historical Baseline	AI-Enabled Target (2027)	Measurement & Control Method
Unplanned Downtime Hours	~320 hours/year	< 270 hours/year (-15%)	SAP Maintenance module tracking
Mean Time Between Failures (MTBF)	1,450 hours	> 1,800 hours	IoT vibration sensor logs
Fouling Analytics Latency	24 hours	Real-time (< 1 minute)	Automated predictive algorithm

3. Comparative Analysis: Shell plc vs. JSC NC KazMunayGas (KMG)

1. **Refinery Modernization**: Shell's Pernis refinery (Netherlands) represents a benchmark in digital refining, utilizing over 50,000 continuous IoT sensors linked to cloud-based machine learning models for real-time energy optimization. KMG's ANPZ has undergone physical modernization (adding catalytic cracking and delayed coking units) but requires digital retrofitting to bridge the operational gap.

2. **Downstream Supply Chain Sensitivity**: Shell can offset a refinery shutdown at Pernis by rerouting product through its global trading network. KMG's three domestic refineries (ANPZ, PNPZ, PKOP) supply 80% of Kazakhstan's fuel market; an unplanned trip at ANPZ immediately threatens national diesel and jet fuel availability.
3. **Capital Expenditure Approach**: Shell invests heavily in custom digital twins for downstream assets. KMG prioritizes targeted, high-ROI retrofits focusing IoT sensor deployment specifically on critical bottleneck assets like heat exchangers and heavy compressors.

4. Strategic Transferability Verdict

- **ADAPT**: Adapt Shell's predictive vibration analysis protocols and thermal fouling models for application on ANPZ's catalytic cracking and hydrotreating units.
- **DELAY**: Delay the implementation of fully automated closed-loop catalyst optimization until sensor density and data network security meet global standards.

5. AI Verification & Level 2 Declaration Log

- **Prompt Used (Gemini 2.5 Pro)**: *"Map the operational process flow for Atyrau Refinery and calculate the exact financial impact of installing IoT sensors."*
- **AI Hallucination Identified**: The AI model claimed that installing 50 IoT sensors would completely eliminate 100% of refinery downtime and increase total refining capacity by 2 million tonnes per year.
- **Human Verification & Correction**: Corrected against technical refinery engineering literature and KMG disclosures. Sensor deployment does not increase nameplate distillation capacity; it improves stream factor. The unrealistic 100% downtime elimination claim was revised to a verified, engineering-feasible 15% reduction in unplanned downtime.

6. Actionable Operational Recommendations & Key Metrics

1. **Sensor Retrofit Plan**: Deploy 120 wireless IoT vibration and temperature sensors across ANPZ heat exchangers and primary pumps by Q3 2027.
2. **Control Room Dashboard**: Integrate real-time thermal fouling alerts into the main ANPZ operator dashboard with clear green/yellow/red risk thresholds.
3. **SOP Update**: Establish mandatory manual engineering inspection protocols whenever the predictive model issues a "Yellow" alert level.

7. Reflection, Peer Review & Capstone Linkage

This process engineering analysis provides the concrete technical evidence required for Attestation A2 (Video Essay Workflow) and supplies the detailed operational schematic for Capstone (Refinery Bottleneck Elimination at ANPZ).